

Notes on Semigroups and Related Structures

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1 Elementary Concepts

1.1 Sets and Relations

1.1.1 Relations

Definition 1.1.1 (Binary relation). Let X be a set. Every subset ρ of the Cartesian product $X \times X$ is called a (*binary*) *relation* on X . If $(x, y) \in \rho$, we also write $x\rho y$, where $x, y \in X$.

Definition 1.1.2 (Domain and range). For a relation $\rho \subseteq X \times X$, the *domain* and *range* of R are defined by

$$\text{dom } \rho = \{x \in X : \exists y \in X \text{ with } (x, y) \in \rho\},$$

and

$$\text{ran } \rho = \{y \in X : \exists x \in X \text{ with } (x, y) \in \rho\}.$$

The word *image* is also used sometimes as a synonym of range.

Notation 1.1.3. The set of all binary relations on X is denoted by $\mathfrak{B}(X)$.

Remark 1.1.4. If $\rho, \tau \in \mathfrak{B}(X)$, then $\rho \subseteq \tau$ means that $x\rho y \Rightarrow x\tau y$ for all $x, y \in X$. We shall often write $\rho \leq \tau$ for $\rho \subseteq \tau$.

Definition 1.1.5 (Complement). The complement of a relation $\rho \in \mathfrak{B}(X)$ is the relation $\bar{\rho} \in \mathfrak{B}(X)$ defined by

$$(x, y) \in \bar{\rho} \iff (x, y) \notin \rho.$$

Definition 1.1.6 (Properties of a relation). A relation $\rho \in \mathfrak{B}(X)$ is called

1. *reflexive*, if $x\rho x$ for all $x \in X$;
2. *symmetric*, if $x\rho y$ implies $y\rho x$;
3. *antisymmetric*, if $x\rho y$ and $y\rho x$ together imply $x = y$;
4. *transitive*, if $x\rho y$ and $y\rho z$ imply $x\rho z$;
5. *right unique*, if ρy and $x\rho z$ imply $y = z$;
6. *left unique*, if $x\rho y$ and $z\rho y$ imply $x = z$;
7. *right total*, if for every $y \in X$ there exists $x \in X$ such that $x\rho y$;
8. *left total*, if for every $x \in X$ there exists $y \in X$ such that $x\rho y$.

The last four notions also apply to relations $\rho \subseteq X \times Y$ for two possibly different sets X and Y .

1.1.2 Mappings

Definition 1.1.7 (Mapping). Let X and Y be sets. A mapping $\varphi : X \rightarrow Y$ is a relation $\varphi \subseteq X \times Y$ which is right unique and left total; that is, for every $x \in X$ there exists exactly one $y \in Y$ such that $(x, y) \in \varphi$.

Remark 1.1.8. As usual, we write $\varphi(x) = y$. Sometimes we also write $x\varphi = y$. The set of all mappings from X to Y is denoted by $\text{Map}(X, Y)$ or Y^X . We also use $\mathcal{T}(X)$ for $\text{Map}(X, X)$. The elements of $\mathcal{T}(X)$ are called *transformations* of X .

Definition 1.1.9 (Injective, surjective, bijective). A mapping $\varphi : X \rightarrow Y$ is called

1. *injective*, if $\varphi(x) = \varphi(x') \Rightarrow x = x'$ for all $x, x' \in X$;
2. *surjective*, if for every $y \in Y$ there exists $x \in X$ such that $\varphi(x) = y$;
3. *bijective*, if it is both injective and surjective.

Equivalently, injectivity means that the relation is left unique, and surjectivity means that the relation is right total.

Remark 1.1.10. The subset of all bijective transformations in $\mathcal{T}(X)$ is denoted by $\mathcal{S}(X)$, and its elements are called *permutations* or *bijections* of X .

Definition 1.1.11 (Identity map). The identity map $\text{id}_X : X \rightarrow X$ is defined by

$$\text{id}_X(x) = x \quad \forall x \in X.$$

Definition 1.1.12 (Constant mapping). For $y \in Y$, the constant mapping $c_y : X \rightarrow Y$ is defined by

$$c_y(x) = y \quad \forall x \in X.$$

Definition 1.1.13 (Restriction). If $\varphi : X \rightarrow Y$ is a mapping and $A \subseteq X$, then the restriction of φ to A is defined by $\varphi|_A : A \rightarrow Y$ such that

$$\varphi|_A(a) = \varphi(a) \quad \forall a \in A.$$

Conversely, we say that φ extends $\varphi|_A$.

Definition 1.1.14 (Partial mapping). A partial mapping $\varphi : X \rightarrow Y$ is defined to be a right unique relation; that is, $\varphi : \text{dom } \varphi \rightarrow Y$ is a mapping, where $\text{dom } \varphi \subseteq X$. The set $\text{dom } \varphi$ is called the domain of φ . The set of all partial mappings on X (from X to X) is denoted by $\mathcal{PT}(X)$.

Remark 1.1.15. The mapping $\varphi : \text{dom } \varphi \rightarrow \text{im } \varphi$ with $\text{dom } \varphi = \text{im } \varphi = \emptyset$ is called the empty mapping.

1.1.3 Order Relations

Definition 1.1.16 (Partial order). A reflexive, antisymmetric, and transitive relation α on a set X is called a *partial ordering* on X , or simply an ordering of X . One writes $x \leq y$ for $(x, y) \in \alpha$, and the pair $(X, \leq)$ is called a *partially ordered set* or a *poset*.

Definition 1.1.17 (Total order). If in addition, a partial ordering has the property that for every $x, y \in X$, either $x \leq y$ or $y \leq x$. In this case, $\leq$ is called a *total order*. Then $(X, \leq)$ is also called a *linearly ordered set* or a *chain*.

Definition 1.1.18 (Maximal and minimal elements). Let X be a poset. An element $w \in X$ is called a *maximal* (respectively, *minimal*) element of $(X, \leq)$ if $w \leq u$ (respectively, $u \leq w$) implies $u = w$ for every $u \in X$.

Definition 1.1.19 (Minimal condition and well-ordering). A poset $(X, \leq)$ satisfies the *minimal condition* if every non-empty subset of X has a minimal element. In that case, we also say that $(X, \leq)$ is *well ordered*.

Definition 1.1.20 (Lower bound and meet). Let Y be a non-empty subset of a poset X . An element $c \in X$ is called a *lower bound* for Y if $c \leq y$ for all $y \in Y$. An element $u \in X$ is called the *greatest lower bound* or *meet* of Y if u is a lower bound for Y and $z \leq u$ for every lower bound z of Y . The greatest lower bound of Y is denoted by $\bigwedge y : y \in Y$. If $Y = \{a, b\}$, we also write $a \wedge b$.

Definition 1.1.21 (Lower semilattice). If $(X, \leq)$ is such that $a \wedge b$ exists for every pair $a, b \in X$, then $(X, \leq)$ is called a *lower semilattice*. If, in addition, $\bigwedge \{y : y \in Y\}$ exists for every non-empty subset $Y \subseteq X$, then $(X, \leq)$ is called a *complete lower semilattice*.

Definition 1.1.22 (Upper bound and join). Similarly, for a non-empty subset $Y \subseteq X$, an element $c \in X$ is called an *upper bound* for Y if $y \leq c$ for all $y \in Y$. An element $v \in X$ is called the *least upper bound* or *join* of Y if v is an upper bound for Y and $v \leq z$ for every upper bound z of Y . The least upper bound of Y is denoted by $\bigvee \{y : y \in Y\}$. If $Y = \{a, b\}$, we also write $a \vee b$.

Definition 1.1.23 (Upper semilattice). If $(X, \leq)$ is such that $a \vee b$ exists for every pair $a, b \in X$, then $(X, \leq)$ is called an *upper semilattice*. If, in addition, $\bigvee y : y \in Y$ exists for every non-empty subset $Y \subseteq X$, then $(X, \leq)$ is called a *complete upper semilattice*.

Definition 1.1.24 (Lattice). If a poset is both a (complete) lower semilattice and a (complete) upper semilattice then it is called a (complete) lattice. In this case we use the notation $(X, \leq, \wedge, \vee)$.

Remark 1.1.25. The set $\mathfrak{B}(X)$ of all binary relations on a set X is a poset under inclusion; that is, $\rho \leq \tau$ if and only if $\rho \subseteq \tau$, for $\rho, \tau \in \mathfrak{B}(X)$. Moreover, it is a lattice with

$$\rho \wedge \tau = \rho \cap \tau \quad \text{and} \quad \rho \vee \tau = \rho \cup \tau,$$

which is called the lattice of relations on X and is denoted by $(\mathfrak{B}(X), \leq, \wedge, \vee)$.

Lemma 1.1.1 (Zorn's lemma). *If $(X, \leq)$ is a non-empty poset such that every chain of elements has an upper bound in X , then X has at least one maximal element.*

Theorem 1.1.2 (Zermelo's theorem on well-ordering). *Every non-empty set can be well ordered.*

1.1.4 Equivalence relations and partitions

Definition 1.1.26. An *equivalence* or *equivalence relation* on X is a reflexive, symmetric and transitive relation on X .

Special equivalence relations on X are $\text{id}_X = \{(x, x) \mid x \in X\}$, the *diagonal relation* on X , also denoted by Δ_X , and the *total relation* $X \times X$, also denoted by ∇_X .

Definition 1.1.27. Let $K \subseteq X$ be a subset. Define the *Rees equivalence* ρ_K by setting

$$x \rho_K x' \iff x, x' \in K \text{ or } x = x'.$$

We denote the set of all equivalences on a set X by $\mathcal{E}(X)$. As a subset of $\mathfrak{B}(X)$ it is partially ordered by inclusion.

We call the intersection of all equivalences containing the relation $\rho \in \mathcal{B}(X)$ the *equivalence on X generated by ρ* or the *equivalence closure* of ρ , and denote it by ρ^e .

If $\{\rho_i \in \mathcal{E}(X) \mid i \in I\}$ is a non-empty set of equivalences on X , then $\bigcap_{i \in I} \rho_i$ is again an equivalence on X .

If $\rho \in \mathcal{B}(X)$, then the intersection of all equivalences containing ρ is the unique minimal equivalence on X containing ρ .

In fact $\mathcal{E}(X)$ is a lattice. If $\rho, \sigma \in \mathcal{E}(X)$, then their greatest lower bound $\rho \wedge \sigma$ is $\rho \cap \sigma$ and their least upper bound $\rho \vee \sigma$ is $(\rho \cup \sigma)^e$.

Definition 1.1.28. For $\rho \in \mathcal{E}(X)$ and $A \subseteq X$, set

$$\rho(A) := \{b \in X \mid \text{there exists } a \in A \text{ with } (a, b) \in \rho\}.$$

For $A = \{a\}$ we write instead of $\rho(a)$ also $[a]_\rho$ or just $[a]$, and call this set the ρ -class of a or the class of a (with respect to ρ).

Let $\rho, \sigma \in \mathcal{E}(X)$. Then $\rho \subseteq \sigma$ if and only if every σ -class is a union of ρ -classes.

Definition 1.1.29. A family of subsets X_i , $i \in I$, of a set X is called a *partition* of X if the subsets X_i are pairwise disjoint, non-empty, and their union is X .

If ρ is an equivalence relation on X , then the set of equivalence classes

$$\pi(\rho) = \{[x]_\rho \mid x \in X\}$$

is the partition of X corresponding to ρ .

On the other hand, if $\pi = \{X_i \mid i \in I\}$ is a partition of X into subsets X_i , then the relation

$$\rho(\pi) = \{(x, y) \in X \times X \mid \text{there exists } i \in I \text{ such that } x, y \in X_i\}$$

is an equivalence relation on X . Note that

$$\rho(\pi(\rho)) = \rho, \quad \pi(\rho(\pi)) = \pi.$$

The set of classes with respect to an equivalence relation ρ is called the *factor set* of X with respect to ρ and denoted by X/ρ . The mapping

$$\pi_\rho : X \rightarrow X/\rho, \quad x \mapsto [x]_\rho$$

is called the *canonical surjection* with respect to ρ .

For any $\rho \in \mathfrak{B}(X)$, let

$$\begin{aligned}\rho^R &= \bigcap \{ \sigma \in B(X) \mid \rho \subseteq \sigma, \sigma \text{ is reflexive} \}, \\ \rho^S &= \bigcap \{ \sigma \in B(X) \mid \rho \subseteq \sigma, \sigma \text{ is symmetric} \}.\end{aligned}$$

There is at least one element $\sigma \in \mathfrak{B}(X)$ satisfying the condition in each of the collections above, namely $\sigma = X \times X$. Furthermore, since every such σ contains ρ , the intersections ρ^R and ρ^S are both non-empty. It is easy to see that

- ρ^R , called the *reflexive closure* of ρ , is the smallest reflexive relation containing ρ ;
- ρ^S , called the *symmetric closure* of ρ , is the smallest symmetric relation containing ρ .

1.1.5 Inverse relation, restriction and composition

Let $\rho \subseteq X \times Y$. The *inverse* (*converse*) relation to ρ is

$$\rho^{-1} = \{ (y, x) \in Y \times X \mid (x, y) \in \rho \}.$$

Similar as for mappings, we now define the *restriction* $\rho|_{X'}$ of a relation $\rho \in B(X)$ for a subset $X' \subseteq X$ by

$$\rho|_{X'} = \rho \cap (X' \times X').$$

The *composition* of relations $\rho, \sigma \in \mathfrak{B}(X)$ is defined as

$$\rho \circ_r \sigma = \{ (x, z) \in X \times X \mid \text{there exists } y \in X \text{ with } x\sigma y \text{ and } y\rho z \},$$

if we start with the right relation σ , and analogously for $\rho \subseteq X \times Y, \sigma \subseteq Y \times Z$ as

$$\rho \circ_r \sigma = \{ (x, z) \in X \times Z \mid \text{there exists } y \in Y \text{ with } x\sigma y \text{ and } y\rho z \}.$$

If we start with the left relation ρ , we define

$$\rho \circ_l \sigma = \{ (x, z) \in X \times X \mid \text{there exists } y \in X \text{ with } x\rho y \text{ and } y\sigma z \},$$

for $\rho, \sigma \in B(X)$, and analogously for $\rho \subseteq X \times Y, \sigma \subseteq Y \times Z$ as

$$\rho \circ_l \sigma = \{ (x, z) \in X \times Z \mid \text{there exists } y \in Y \text{ with } x\rho y \text{ and } y\sigma z \}.$$

Usually, for convenience, we will denote either $\circ_r$ or $\circ_l$ by $\circ$. We will be careful in any situation to indicate the use of either convention.

If $\varphi \in \text{Map}(X, Y)$ and $\psi \in \text{Map}(Y, Z)$, then in the case when mapping symbols are written to the left of the elements on which they act, by $\psi\varphi$ we mean $\psi \circ_r \varphi$, i.e.

$$(\psi\varphi)(x) = \psi(\varphi(x)) \quad \forall x \in X.$$

In the case when mapping symbols are written to the right of the elements on which they act, by $x\varphi\psi$ we mean $\varphi \circ_l \psi$, i.e.

$$(x\varphi)\psi = (x\varphi)\psi \quad \forall x \in X.$$

1.1.6 Transitive and equivalence closure

It is clear that the operation $\circ$ on $\mathfrak{B}(X)$ is associative and thus $(\mathfrak{B}(X), \circ)$ will be one of the examples of monoids (definition in a further section) with the identity element id_X . Sometimes we write $\rho\sigma$ for $\rho \circ \sigma$. We define powers of a relation $\rho \in \mathfrak{B}(X)$ as follows:

$$\rho^0 = id_X, \quad \rho^i = \rho \circ \rho \circ \dots \circ \rho, \text{ } i \text{ times, } i \in \mathbb{N}.$$

Definition 1.1.30. The *transitive closure* of a relation $\rho \in \mathfrak{B}(X)$ is the relation

$$\rho^\infty = \bigcup_{n=1}^{\infty} \rho^n \in \mathfrak{B}(X).$$

Lemma 1.1.3. If $\rho \in \mathfrak{B}(X)$, then ρ^∞ is the smallest transitive relation in $(\mathfrak{B}(X), \leq)$ containing ρ .

Proof. If $(x, y), (y, z) \in \rho^\infty$, then there exist $m, n \in \mathbb{N}$ such that $(x, y) \in \rho^m, (y, z) \in \rho^n$, and

$$(x, z) \in \rho^m \circ \rho^n = \rho^{m+n} \subseteq \rho^\infty.$$

Thus ρ^∞ is a transitive relation and $\rho \subseteq \rho^\infty$. If now σ is a transitive relation on X and $\rho \subseteq \sigma$, then

$$\rho^2 = \rho \circ \rho \subseteq \sigma \circ \sigma \subseteq \sigma.$$

By induction it follows easily that $\rho^n \subseteq \sigma$ for all $n \in \mathbb{N}$, i.e. $\rho^\infty \subseteq \sigma$. □

Theorem 1.1.4. If $\rho \in \mathfrak{B}(X)$, then for the equivalence closure ρ^e we get

$$\rho^e = (\rho \cup \rho^{-1} \cup id_X)^\infty.$$

Proof. It is clear that $\tau = (\rho \cup \rho^{-1} \cup id_X)^\infty$ contains ρ and is transitive and reflexive. The relation $\nu = \rho \cup \rho^{-1} \cup id_X$ is symmetric and therefore $\nu^n = (\nu^{-1})^n = (\nu^n)^{-1}$. Thus, if $(x, y) \in \tau$, then $(x, y) \in \nu^n$ for some $n \in \mathbb{N}$ and therefore $(y, x) \in \nu^n$. Thus $(y, x) \in \tau$, i.e. τ is symmetric. This means that $\tau \in \mathcal{E}(X)$.

If $\sigma \in \mathcal{E}(X)$ and $\rho \subseteq \sigma$, then $id_X \subseteq \sigma, \rho^{-1} \subseteq \sigma^{-1} = \sigma$, and therefore $\nu = \rho \cup \rho^{-1} \cup id_X \subseteq \sigma$. Since $\nu \circ \nu \subseteq \sigma \circ \sigma \subseteq \sigma$, we get $\nu^n \subseteq \sigma$ for all $n \in \mathbb{N}$, and hence $\tau \subseteq \sigma$. □

Corollary 1.1.5. If $\rho \in \mathfrak{B}(X)$, then $(x, y) \in \rho^e$ if and only if $x = y$ or for some $n \in \mathbb{N}$ there exists a sequence of elements $x_0, x_1, x_2, \dots, x_n$ in X such that $x_0 = x, x_n = y$, and for every $i \in \{1, \dots, n\}$ either $(x_{i-1}, x_i) \in \rho$ or $(x_i, x_{i-1}) \in \rho$.

In particular, if ρ and σ are equivalence relations on a set X , then in $\mathcal{E}(X)$ their join $\rho \vee \sigma$ is the relation defined by

$$x(\rho \vee \sigma)y \iff \text{there exist } x_0, x_1, \dots, x_n \in X \text{ such that } x = x_0, y = x_n$$

and

$$(x_{i-1}, x_i) \in \tau_i, \quad \tau_i \in \{\rho, \sigma\}, \quad i \in \{1, \dots, n\}.$$

Proposition 1.1.6. For any $\rho \in \mathfrak{B}(X)$,

1. $\rho^R = \rho \cup id_X$;
2. $\rho^S = \rho \cup \rho^{-1}$;
3. $(\rho^R)^S = (\rho^S)^R = \rho \cup \rho^{-1} \cup id_X$;
4. $\rho^e = \left((\rho^R)^S \right)^\infty$.

1.1.7 Kernel equivalence and Homomorphism Theorem for sets

Definition 1.1.31. Let $\varphi : X \rightarrow Y$ be a mapping. The relation

$$\ker \varphi = \{(x, x') \in X \times X \mid \varphi(x) = \varphi(x')\}$$

is an equivalence on X which is called the *kernel equivalence* or the *kernel* of φ . It follows from the definition of the composition of relations that

$$\ker \varphi = \varphi^{-1} \circ \varphi = \varphi^{-1} \circ_r \varphi.$$

Theorem 1.1.7 (Homomorphism Theorem for sets). *Let $\varphi : X \rightarrow Y$ be a mapping and let ρ be an equivalence on X such that $x\rho x'$ implies $\varphi(x) = \varphi(x')$, i.e. $\rho \leq \ker \varphi$. Then $\varphi' : X/\rho \rightarrow Y$, with $\varphi'([x]_\rho) := \varphi(x)$, $x \in X$, is the unique mapping such that the following diagram is commutative:*

$$\begin{array}{ccc} X & \xrightarrow{\varphi} & Y \\ \pi_\rho \downarrow & \nearrow \varphi' & \\ X/\rho & & \end{array}$$

i.e. such that $\varphi' \pi_\rho = \varphi$, where π_ρ denotes the canonical surjection.

If $\rho = \ker \varphi$, then φ' is injective, and if φ is surjective, then φ' is surjective.

Proof. Define $\varphi' : X/\rho \rightarrow Y$ by $\varphi'([x]_\rho) = \varphi(x)$ for any $x \in X$. Suppose $[x]_\rho = [x']_\rho$ for $x, x' \in X$. Then $(x, x') \in \rho$ and thus $\varphi(x) = \varphi(x')$. Hence φ' is well-defined.

Suppose $\varphi'([x]_\rho) = \varphi'([x']_\rho)$. Then $\varphi(x) = \varphi(x')$ and thus $(x, x') \in \ker \varphi$.

If $\rho = \ker \varphi$, then $[x]_\rho = [x']_\rho$ and thus φ' is injective. □

1.1.8 Congruences

Definition 1.1.32. A relation ρ on X is called

- *left compatible* if for all $x, y, z \in X$,

$$(x, y) \in \rho \implies (zx, zy) \in \rho;$$

- *right compatible* if for all $x, y, z \in X$,

$$(x, y) \in \rho \implies (xz, yz) \in \rho;$$

- *compatible* if for all $x, y, s, t \in X$,

$$(x, y) \in \rho \text{ and } (s, t) \in \rho \implies (xs, yt) \in \rho.$$

A left compatible equivalence relation is called a *left congruence*; a right compatible equivalence relation is called a *right congruence*; and a compatible equivalence relation is called a *congruence*.

Proposition 1.1.8. *A relation ρ on X is a congruence if and only if it is both a left and a right congruence.*

1 Elementary Concepts

Proof. Suppose ρ is both a left and a right congruence. Let $x, y, z, t \in X$ be such that $(x, y) \in \rho$ and $(z, t) \in \rho$. Since ρ is a right congruence, $(xz, yz) \in \rho$, and since ρ is a left congruence, $(yz, yt) \in \rho$. By transitivity of ρ , it follows that $(xz, yt) \in \rho$. Hence ρ is a congruence.

Conversely, suppose ρ is a congruence. Let $x, y \in X$ be such that $(x, y) \in \rho$, and let $z \in X$. Since ρ is reflexive, $(z, z) \in \rho$. By compatibility, $(zx, zy) \in \rho$. Thus ρ is left compatible. Similarly, $(xz, yz) \in \rho$, so ρ is right compatible. Hence ρ is both a left and a right congruence. $\square$

For any $\rho \in \mathfrak{B}(X)$, let

$$\rho^c = \bigcap \{ \sigma \in \mathfrak{B}(X) \mid \rho \subseteq \sigma, \sigma \text{ is left and right compatible} \}.$$

Then ρ^c is the smallest left and right compatible relation containing ρ . Also, let

$$\rho^\# = \bigcap \{ \sigma \in \mathfrak{B}(X) \mid \rho \subseteq \sigma, \sigma \text{ is a congruence} \}.$$

Then ρ^c , called the congruence generated by ρ , is the smallest congruence containing ρ .

Proposition 1.1.9. For any $\rho, \sigma \in \mathfrak{B}(X)$,

1. $(\rho \cup \sigma)^c = \rho^c \cup \sigma^c$;
2. $(\rho^{-1})^c = (\rho^c)^{-1}$.

Proposition 1.1.10. For any $\rho \in \mathfrak{B}(X)$, we have

$$\rho^\# = (\rho^c)^e.$$

Proof. By definition, $(\rho^c)^e$ is an equivalence relation containing ρ^c , and hence containing ρ . Notice that

$$\begin{aligned} (\rho^c)^e &= (\rho^c \cup (\rho^c)^{-1} \cup id_X)^\infty \\ &= ((\rho \cup \rho^{-1} \cup id_X)^c)^\infty \\ &= \bigcup_{n=0}^\infty ((\rho \cup \rho^{-1} \cup id_X)^c)^n \end{aligned}$$

Now, for all $x, y, z \in X$,

$$\begin{aligned} &(x, y) \in (\rho^c)^e \\ \implies &(x, y) \in \bigcup_{n=0}^\infty ((\rho \cup \rho^{-1} \cup id_X)^c)^n \\ \implies &\exists n \in \mathbb{N} \text{ such that } (x, y) \in ((\rho \cup \rho^{-1} \cup id_X)^c)^n \\ \implies &\exists x_0, x_1, \dots, x_n \text{ such that } x = x_0 \text{ and } y = x_n \text{ and } (\forall i) (x_i, x_{i+1}) \in (\rho \cup \rho^{-1} \cup id_X)^c \\ \implies &zx = zx_0, zx_n = zy \text{ and } (\forall i) (zx_i, zx_{i+1}) \in (\rho \cup \rho^{-1} \cup id_X)^c \\ &\quad (\text{since } (\rho \cup \rho^{-1} \cup id_X)^c \text{ is left and right congruence}) \\ \implies &(zx, zy) \in ((\rho \cup \rho^{-1} \cup id_X)^c)^n \\ \implies &(zx, zy) \in \bigcup_{n=0}^\infty ((\rho \cup \rho^{-1} \cup id_X)^c)^n \\ \implies &(zx, zy) \in (\rho^c)^e. \end{aligned}$$

Therefore, $(\rho^c)^e$ is left compatible. Similarly, $(\rho^c)^e$ is right compatible. Hence $(\rho^c)^e$ is a congruence containing ρ .

Finally, let τ be any congruence containing ρ . Then τ is left and right compatible, so $\rho^c \subseteq \tau$. Since τ is an equivalence relation, it must contain $(\rho^c)^e$. Hence $(\rho^c)^e \subseteq \tau$. Therefore, $(\rho^c)^e$ is the smallest congruence containing ρ , and therefore $\rho^\# = (\rho^c)^e$. Consequently, $\rho^\# = (\rho^c)^e$ $\square$

We denote by $\mathcal{C}(X)$ the set of all congruences on X . Then $\mathcal{C}(X)$, as a subset of $\mathcal{E}(X)$ and $\mathcal{B}(X)$, is partially ordered by inclusion.

In fact, $\mathcal{C}(X)$ forms a lattice. For $\rho, \sigma \in \mathcal{C}(X)$, we have

$$\rho \wedge \sigma = \rho \cap \sigma \quad \text{and} \quad \rho \vee \sigma = (\rho \cup \sigma)^\#.$$

Also,

$$\begin{aligned} (\rho \cup \sigma)^\# &= ((\rho \cup \sigma)^c)^e \\ &= (\rho^c \cup \sigma^c)^e \\ &= (\rho \cup \sigma)^e \end{aligned}$$

so there is no ambiguity in writing $\rho \vee \sigma$ whether one means the join $(\rho \vee \sigma)^e$ in the lattice $\mathcal{E}(X)$ or the join $(\rho \vee \sigma)^\#$ in the lattice $\mathcal{C}(X)$.

Proposition 1.1.11. *Let $\rho, \sigma \in \mathcal{C}(X)$. Then*

$$\rho \vee \sigma = (\rho \circ \sigma)^\infty.$$

Proof. Let $\rho, \sigma \in \mathcal{C}(X)$. Since $\rho \vee \sigma$ contains both ρ and σ , it follows that

$$\rho \circ \sigma \subseteq (\rho \cup \sigma) \circ (\rho \cup \sigma) = (\rho \vee \sigma)^2.$$

More generally, $(\rho \circ \sigma)^n \subseteq (\rho \cup \sigma)^{2n}$. Thus,

$$(\rho \circ \sigma)^\infty = \bigcup_{n=0}^{\infty} (\rho \circ \sigma)^n \subseteq \bigcup_{n=0}^{\infty} (\rho \cup \sigma)^{2n} = (\rho \cup \sigma)^\infty.$$

On the other hand, $\rho \circ \sigma \supseteq \rho \circ id_X = \rho$ (since σ is reflexive), and $\rho \circ \sigma \supseteq id_X \circ \sigma = \sigma$ (since ρ is reflexive). Thus, $\rho \cup \sigma \subseteq \rho \circ \sigma$. Hence, $(\rho \cup \sigma)^\infty \subseteq (\rho \circ \sigma)^\infty$.

Consequently, $(\rho \cup \sigma)^\infty = (\rho \circ \sigma)^\infty$.

Then,

$$\begin{aligned} \rho \vee \sigma &= (\rho \cup \sigma)^e = \left(((\rho \cup \sigma)^R)^S \right)^\infty \\ &= ((\rho \cup \sigma) \cup (\rho \cup \sigma)^{-1} \cup id_X)^\infty \\ &= (\rho \cup \sigma \cup \rho^{-1} \cup \sigma^{-1} \cup id_X)^\infty \\ &= (\rho \cup \sigma \cup id_X)^\infty \\ &= (\rho \cup \sigma)^\infty = (\rho \circ \sigma)^\infty. \end{aligned}$$

This completes the proof. □

Proposition 1.1.12. *Let $\rho, \sigma \in \mathcal{C}(X)$. If $\rho \circ \sigma = \sigma \circ \rho$, then*

$$\rho \vee \sigma = \rho \circ \sigma.$$

Proof. Suppose $\rho \circ \sigma = \sigma \circ \rho$. Then

$$(\rho \circ \sigma)^2 = \rho \circ \sigma \circ \rho \circ \sigma = \rho \circ \rho \circ \sigma \circ \sigma = \rho \circ \sigma,$$

since ρ and σ are transitive.

Hence, $(\rho \circ \sigma)^n = \rho \circ \sigma$ for all $n \in \mathbb{N}$, and therefore

$$\rho \circ \sigma = (\rho \circ \sigma)^\infty = \bigcup_{n \geq 0} (\rho \circ \sigma)^n = \rho \circ \sigma.$$

So $\rho \vee \sigma = \rho \circ \sigma$. □

Lemma 1.1.13. *If $\rho = \cap_i \rho_i$ (an intersection of congruences, or even just equivalence relations on a set X), then for any $x \in X$*

$$[x]_\rho = \cap_i [x]_{\rho_i}.$$

Proof. ; For any $y \in X$

$$x \in [x]_\rho \iff (x, y) \in \rho \iff (x, y) \in \rho_i \quad \forall i \iff y \in [x]_{\rho_i} \quad \forall i \iff y \in \cap_i [x]_{\rho_i}$$

□

1.2 Introductory Algebraic structures

1.2.1 Groupoids, Identities, and Zeros

Definition 1.2.1. A *groupoid* is a non-empty set A together with a binary operation

$$A \times A \rightarrow A, \quad (a, a') \mapsto aa'.$$

The operation is usually called the *multiplication* of the groupoid.

Remark 1.2.2. Let A be a groupoid. For non-empty subsets $B, C \subseteq A$, define

$$BC = \{bc \mid b \in B, c \in C\}.$$

We write $B^2 = BB$, and similarly for higher products.

Definition 1.2.3. Let S be a groupoid. An element $e \in S$ is called

- a *left identity* if $es = s$ for all $s \in S$;
- a *right identity* if $se = s$ for all $s \in S$;
- an *identity* if $es = s = se$ for all $s \in S$.

An identity is also called a *neutral element*.

Proposition 1.2.1. *If a groupoid S has both a left and a right identity, then they coincide. Hence a groupoid has at most one identity.*

Proof. Let e' be a left identity and e'' a right identity. Then

$$e'' = e'e'' = e'.$$

□

We denote the identity (when it exists) by 1 or more precisely by 1_S .

Definition 1.2.4. Let S be a groupoid with identity.

- $s \in S$ is *left invertible* if there exists $t \in S$ such that $ts = 1$.
- s is *right invertible* if there exists $t \in S$ such that $st = 1$.
- s is *invertible* if there exists $t \in S$ satisfying $ts = st = 1$.

In third case, such an element t is called an inverse of s , and is denoted by s^{-1} whenever it is unique.

Definition 1.2.5. Let S be a groupoid. An element $z \in S$ is called

- a *left zero* if $zs = z$ for every $s \in S$;
- a *right zero* if $sz = z$ for every $s \in S$;
- a *zero* if $zs = z = sz$ for every $s \in S$.

A zero is usually denoted by 0.

Exercise 1.2.2. Show that if a groupoid has both a left zero and a right zero, then they coincide. Hence a groupoid has at most one zero.

1.2.2 Semigroups, monoids and groups

Definition 1.2.6. A *semigroup* is a groupoid whose multiplication is associative. A semigroup with an identity element is called a *monoid*.

Proposition 1.2.3. If an element of a monoid has both a left and a right inverse, then they coincide.

Proof. Suppose $ts = 1$ and $st' = 1$. Then

$$t' = (ts)t' = t(st') = t. \quad \square$$

Corollary 1.2.4. Every element of a monoid has at most one inverse.

Definition 1.2.7. A *group* G is a monoid in which every element is invertible.

For any $a, b \in G$,

$$(ab)^{-1} = b^{-1}a^{-1}, \quad (a^{-1})^{-1} = a.$$

Exercise 1.2.5. Suppose S is a semigroup with an element e satisfying $es = s$ for every $s \in S$, and suppose every $s \in S$ has an element $t \in S$ such that $ts = e$. Show that S is a group.

Remark 1.2.8 (Free semigroup and free monoid). Let X be a non-empty set, called an *alphabet*. Define

$$X^+ = \{x_1x_2 \cdots x_n \mid n \in \mathbb{N}, x_i \in X, 1 \leq i \leq n\},$$

the set of all finite non-empty words over X .

Two words

$$x_1x_2 \cdots x_n, \quad y_1y_2 \cdots y_m$$

are equal if and only if

$$n = m \quad \text{and} \quad x_i = y_i \quad \text{for all } 1 \leq i \leq n.$$

Define the multiplication (concatenation) of words by

$$(x_1x_2 \cdots x_n)(y_1y_2 \cdots y_m) = x_1x_2 \cdots x_ny_1y_2 \cdots y_m.$$

With this operation, X^+ forms a semigroup, called the *free semigroup* on X , or the *free semigroup with basis X* .

Now let

$$X^* = X^+ \cup \{\varepsilon\},$$

where ε denotes the empty word. Then X^* is a monoid under concatenation, with identity element ε . It is called the *free monoid* on X , or the *free monoid with basis X* .

1.2.3 Examples

Example 1.2.9 (Groupoids). 1. $(\mathbb{Z}, -)$.

2. $(\mathbb{Q} \setminus \{0\}, /)$.

3. $(\mathbb{N}, \uparrow)$, where $n \uparrow m = n^m$.

4. The vector product $(\mathbb{R}^3, \times)$.

Example 1.2.10 (Semigroups and monoids). 1. If $(R, +, \cdot)$ is a ring, then $(R, +)$ is a group and $(R, \cdot)$ is a semigroup.

2. $(\mathbb{N}, \cdot)$ is a monoid.

3. $(\mathbb{N}, +)$ is a semigroup.

4. $(\mathcal{T}(X), \circ)$, the full transformation monoid.

5. $(\mathcal{PT}(X), \circ)$, the monoid of partial transformations.

6. $(\mathfrak{B}(X), \circ)$, the semigroup of binary relations.

7. The free semigroup X^+ .

8. The free monoid X^* .

9. $(\mathbb{Z}, \gcd)$.

10. **The bicyclic monoid:**

Let $B = \mathbb{N}_0 \times \mathbb{N}_0$, where $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. Define a binary operation on B by

$$(m, n)(p, q) = (m - n + \max\{n, p\}, q - p + \max\{n, p\}).$$

Equivalently,

$$(m, n)(p, q) = \begin{cases} (m - n + p, q), & \text{if } p \geq n, \\ (m, n - p + q), & \text{if } p \leq n. \end{cases}$$

Then $(B, \cdot)$ is a monoid, called the *bicyclic monoid*, with identity element $(0, 0)$.

Example 1.2.11 (Groups). 1. $(\mathbb{Z}, +)$.

2. $(\mathbb{Q}, +)$.

3. $(\mathbb{Q} \setminus \{0\}, \cdot)$.

4. The residue class group $(\mathbb{Z}_r, +)$.

5. The symmetric group $S(X)$, or S_n when $|X| = n$.

2 The Semigroup

2.1 Semigroups and Monoids

Throughout this chapter, S will denote a semigroup with operation $\circ$. Formally, we write $(S, \circ)$ to indicate that we are considering the set S with the operation $\circ$, but we will only do this when we need to distinguish a particular operation. Unless we need to distinguish between different operations, we will often write xy instead of $x \circ y$ (where $x, y \in S$), and we will call the operation multiplication and the element xy (i.e. the result of applying the operation to x and y) the product of x and y .

Since the operation is associative, there is no ambiguity in the product $s_1 s_2 \cdots s_n$ (where each $s_i \in S$): the product is the same regardless of how we insert the brackets.

Any group is also a semigroup. The most familiar example of a semigroup that is not a group is the set of natural numbers $\mathbb{N}$ under the operation of addition. This is not a group since it contains no identity. (For our purposes, $\mathbb{N} = \{1, 2, 3, \dots\}$ does not include 0.)

Example 2.1.1. The trivial semigroup contains only one element e , with multiplication obviously defined by $ee = e$. Since e is (trivially) an identity, this semigroup is also called the trivial monoid.

2.1.1 Basic Concepts

Definition 2.1.2. A null semigroup is a semigroup with a zero z in which the product of any two elements is z . It is easy to see that this multiplication is associative. Notice that we can define a null semigroup on any set by choosing some element z and defining all products to be z .

Definition 2.1.3. If every element of S is a left zero (that is, $xy = x$ for all $x, y \in S$), then S is a left zero semigroup. If every element of S is a right zero (that is, $xy = y$ for all $x, y \in S$), then S is a right zero semigroup. We can define a left zero semigroup on any set X by defining the multiplication $xy = x$ for all $x, y \in X$; it is easy to see that this multiplication is associative.

If a semigroup does not contain identity (zero element), we can always add a new element to act as an identity (zero element). To see this, let 1 be a new element not in the semigroup S . Extend the multiplication on S to $S \cup \{1\}$ by $1x = x1 = x$ for all $x \in S$ and $11 = 1$. It is easy to prove that this extended multiplication is associative. Then $S \cup \{1\}$ is a monoid with identity 1 . Similarly, let 0 be a new element not in S and extend the multiplication on S to $S \cup \{0\}$ by $0x = x0 = 00 = 0$ for all $x \in S$. Again, this extended multiplication is associative. Then $S \cup \{0\}$ is a semigroup with zero 0 .

For any semigroup S , define

$$S^1 = \begin{cases} S, & \text{if } S \text{ has an identity,} \\ S \cup \{1\}, & \text{otherwise,} \end{cases} \quad S^0 = \begin{cases} S, & \text{if } S \text{ has a zero,} \\ S \cup \{0\}, & \text{otherwise.} \end{cases}$$

2 The Semigroup

The semigroups S^1 and S^0 are called, respectively, the monoid obtained by adjoining an identity to S if necessary and the semigroup obtained by adjoining a zero to S if necessary.

Throughout this chapter, mappings are written on the right and composed left to right. To clarify: let $\varphi : X \rightarrow Y$ and $\psi : Y \rightarrow Z$ be maps. The result of applying φ to an element x of X is denoted $x\varphi$. The composition of φ and ψ is denoted $\varphi\psi$ or simply $\varphi\psi$, and is a map from X to Z with $x(\varphi\psi) = (x\varphi)\psi$ for all $x \in X$.

Definition 2.1.4 (Cartesian Product). Let $X = \{X_i : i \in I\}$ be a collection of sets. Informally, the Cartesian product $\prod_{i \in I} X_i$ of the sets in X is the set of tuples with $|I|$ components, where, for each $i \in I$, the i -th component is an element of X_i . More formally, the Cartesian product $\prod_{i \in I} X_i$ is the set of mappings σ from I to $\bigcup_{i \in I} X_i$ such that $i\sigma \in X_i$ for each $i \in I$.

We think the map σ as a tuple with i -th component $i\sigma$. We will use both mapping notation and (especially when the index set I is finite) tuple notation for elements of Cartesian products. When $I = \{1, \dots, n\}$ is finite, we write $X_1 \times \dots \times X_n$ for $\prod_{i \in I} X_i$. When the sets X_i are all equal to a set X , that is, when we consider products of $|I|$ copies of the set X , we write X^I for $\prod_{i \in I} X_i$.

Definition 2.1.5. Let $S = \{S_i : i \in I\}$ be a collection of semigroups. The direct product of the semigroups in S is their Cartesian product $\prod_{i \in I} S_i$ with componentwise multiplication: $(i)(s)(i)(t) = (i)(s)(i)t$, or, using tuple notation, if $(\dots, s_i, \dots), (\dots, t_i, \dots) \in \prod_{i \in I} S_i$, then

$$(\dots, s_i, \dots)(\dots, t_i, \dots) = (\dots, s_i t_i, \dots).$$

It is easy to prove that this componentwise multiplication is associative, and so the direct product is itself a semigroup.

For $x \in S$ and $n \in \mathbb{N}$, define

$$x^n = \underbrace{x \cdots x}_{n \text{ times}}.$$

In general, x^n is only defined for positive n . If S is a monoid, define $x^0 = 1_S$. An element x of S is *idempotent* if $x^2 = x$. The set of idempotents of S is denoted $E(S)$.

For any subsets X and Y of S , define $XY = \{xy : x \in X, y \in Y\}$. Write xY for $\{x\}Y$ and XY for $X\{y\}$. Since multiplication in S is associative, so is this product of subsets: for subsets X, Y, Z of S , we have $X(YZ) = (XY)Z$. By analogy with the above, for $X \subseteq S$ and $n \in \mathbb{N}$, define

$$X^n = \underbrace{X \cdots X}_{n \text{ times}}.$$

Definition 2.1.6 (Nilpotent Semigroup). The semigroup S is nilpotent if it contains a zero and there exists some $n \in \mathbb{N}$ such that $S^n = \{0\}$.

Definition 2.1.7. The semigroup S is a nilsemigroup if it contains a zero and for every $x \in S$, there exists some $n \in \mathbb{N}$ such that $x^n = \{0\}$.

Definition 2.1.8 (Cancellative Semigroup). The semigroup S is left cancellative if

$$(\forall x, y, z \in S)(zx = zy \Rightarrow x = y);$$

right cancellative if

$$(\forall x, y, z \in S)(xz = yz \Rightarrow x = y);$$

and cancellative if it is both left and right cancellative.

Note 2.1.9. A non-trivial semigroup with zero cannot be cancellative.

Definition 2.1.10 (Commutativity). The semigroup S is commutative if $xy = yx$ for all $x, y \in S$. For instance, $(\mathbb{Z}, +)$ and $(\mathbb{Z}, \cdot)$ are both commutative.

A non-trivial left zero semigroup is not commutative.

2.1.2 Regular Semigroups

Let $x \in S$. If there is an element $y \in S$ such that $xyx = x$, then the element x is *regular*. Notice that in this case, $(xy)^2 = xyxy = x(yx)y = xy$ and $(yx)^2 = yxyx = y(xy)x = yx$ and hence xy and yx are idempotent elements of S .

Definition 2.1.11 (Regular Semigroup). If every element of S is regular, then S is a regular semigroup.

Now we define the inverse of an element of a semigroup S in general.

Definition 2.1.12. Let $x \in S$. An element $x' \in S$ is called as inverse of x if $x = xx'x$ and $x' = x'xx'$.

Notice that this is entirely different from the notion of left/right inverses above. We will never use *inverse* (on its own) to refer to a left or right inverse.

Proposition 2.1.1. Let $x \in S$. Then x has an inverse if and only if x is regular.

Proof. Obviously if x has an inverse, then it is regular. So suppose x is regular. Then there exists $y \in S$ such that $xyx = x$. Let $x' = yxy$. Then

$$xx'x = x(yxy)x = (xyx)yxyx = xyx = x$$

and

$$x'xx' = (yxy)x(yxy) = y(xy x)yxy = yxy = x',$$

so x' is an inverse of x . □

In the proof of Proposition 2.1.1, the element y might not be an inverse of x ; for example, let S be a semigroup with a zero and let $x = 0$ and $y \neq 0$.

An element x can have more than one inverse. The set of inverses of x is denoted $V(x)$. Notice also that a zero of a semigroup has an inverse, namely itself. In general, if $e \in S$ is idempotent, then $e^2 = e$ and so e is an inverse of itself. In particular, every idempotent is regular.

Example 2.1.13. (a) Let $U = \{0, \dots, k\}$ for some $k > 0$. Define $m\Delta n = \min\{m, n\}$. It is easy to see that Δ is associative, and so (U, Δ) is a semigroup. Notice that $0\Delta m = m\Delta 0 = 0$ and $k\Delta m = m\Delta k = m$ for all $m \in U$. Hence U has zero 0 and identity k . Furthermore, $m\Delta m = m$ for all $m \in U$, so every element of U is idempotent. Finally, $m\Delta n = n\Delta m$ for all $m, n \in U$, so U is commutative.

(b) Define a similar associative operation Δ on $\mathbb{N}_0 \cup \{0\} = \{0, 1, 2, \dots\}$ by

$$m\Delta n = \min\{m, n\}.$$

Then $(\mathbb{N}, \Delta)$ has a zero 0 but has no identity. It is commutative and all its elements are idempotents.

(c) Consider the set of all 2×2 integer matrices:

$$M_2(\mathbb{Z}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{Z} \right\}.$$

2 The Semigroup

With the usual matrix multiplication, $M_2(\mathbb{Z})$ is a monoid with identity

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

and zero

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

It is easy to see that $M_2(\mathbb{Z})$ is not commutative, that not all of its elements are idempotents. Since $M_2(\mathbb{Z})$ contains a zero, it is not cancellative.

- (d) Now let V be the set of all 2×2 integer matrices with non-zero determinant. Again, V is a monoid. Let $P, Q, R \in V$. Suppose $RP = RQ$. Since $\det R \neq 0$, the matrix R has an inverse $R^{-1} \in M_2(\mathbb{Q})$. Hence V is left-cancellative. Similarly, it is right-cancellative and therefore cancellative.

Definition 2.1.14. Let L be a left zero semigroup and R a right zero semigroup. Let $B = L \times R$. This semigroup is called as $|L| \times |R|$ rectangular band, or simply a *rectangular band*.

For $(\ell_1, r_1), (\ell_2, r_2) \in B$, we have

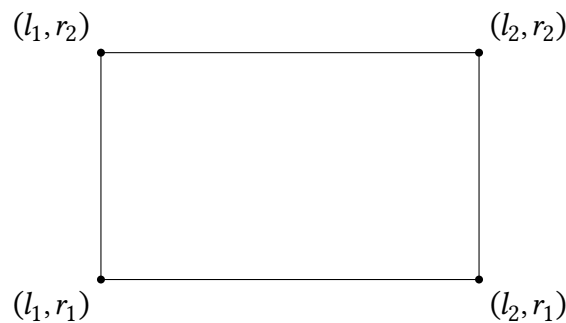
$$(\ell_1, r_1)(\ell_2, r_2) = (\ell_1, r_2),$$

since (in particular) ℓ_1 is a left zero and r_2 is a right zero. Thus every element of B is idempotent, since $(\ell, r)(\ell, r) = (\ell, r)$ for all $(\ell, r) \in B$. Furthermore, for any $(\ell_1, r_1), (\ell_2, r_2) \in B$, we have

$$(\ell_1, r_1)(\ell_2, r_2)(\ell_1, r_1) = (\ell_1 \ell_2 \ell_1, r_1 r_2 r_1) = (\ell_1, r_1),$$

$$(\ell_2, r_2)(\ell_1, r_1)(\ell_2, r_2) = (\ell_2 \ell_1 \ell_2, r_2 r_1 r_2) = (\ell_2, r_2).$$

Hence (ℓ_2, r_2) is an inverse of (ℓ_1, r_1) . Thus every element is an inverse of every element. The name *rectangular band* comes from the following geometric interpretation of the multiplication. To find the product $(\ell_1, r_1)(\ell_2, r_2)$, take the rectangle two of whose vertices have coordinates (ℓ_1, r_1) and (ℓ_2, r_2) . Then the product is the vertex vertically in line with (ℓ_1, r_1) and horizontally in line with (ℓ_2, r_2) , as shown in Figure 2.1.2.



2.1.3 Subsemigroups

Definition 2.1.15 (Subsemigroup). A non-empty subset T of S is a subsemigroup if it is closed under multiplication; that is, if $TT \subseteq T$. A proper subsemigroup is any subsemigroup except S itself. A submonoid is a subsemigroup that happens to be a monoid. A subgroup is a subsemigroup that happens to be a group.

Proposition 2.1.2. *The set of invertible elements of a monoid forms a subgroup.*

Proof. Let T be the set of invertible elements of a monoid M . Note that T is non-empty since $1 \in T$. Let $x, y \in T$. Then since x and y are invertible we have

$$y^{-1}x^{-1}xy = y^{-1}y = 1 \quad \text{and} \quad xyy^{-1}x^{-1} = xx^{-1} = 1.$$

Hence xy is invertible and so lies in T . Hence T is a subsemigroup of M . Furthermore, $1 \in T$ is also an identity for T , and so T is a submonoid of M . Finally, all elements of T are invertible, and so T is a subgroup of M . $\square$

Proposition 2.1.2 makes following definition justifiable:

Definition 2.1.16 (Group of units). The set of invertible elements of a monoid is called its group of units.

Lemma 2.1.3. *Let G be a subset of S . Then $gG = Gg = G$ for all $g \in G$ if and only if G is a subgroup of S .*

Proof. Notice first that if G is a subgroup and $g \in G$, then

$$G = gg^{-1}G \subseteq gG \subseteq G,$$

so $G = gG$. Similarly $G = Gg$.

For the converse, suppose that $gG = Gg = G$ for all $g \in G$. Let $g, h \in G$. Then $gh \in gG = G$, and so G is a subsemigroup.

Since $g \in G = Gg$, there exists $e \in G$ such that $eg = g$. Since $h \in G = gG$, we have $h = gx$ for some $x \in G$, and then

$$eh = egx = gx = h.$$

Thus e is a left identity for G . Similarly G contains a right identity f , and so $e = f$ is an identity for G by Proposition 1.2.1. So G is a submonoid with identity 1_G .

Finally, since $1_G \in gG = Gg$, the element g is right and left invertible, and its right and left inverses coincide by Proposition 1.2.3. Since $g \in G$ was arbitrary, G is a subgroup. $\square$

2.1.4 Generators

Let $\mathcal{T} = \{T_i : i \in I\}$ be a collection of subsemigroups of S . Then if their intersection $\bigcap T = \bigcap_{i \in I} T_i$ is non-empty, it is also a subsemigroup.

Definition 2.1.17 (Generating a subsemigroup). Let $X \subseteq S$, and let $\mathcal{T}$ be the collection of subsemigroups of S that contain X . The collection $\mathcal{T}$ has at least one member, namely the semigroup S itself, and every subsemigroup in $\mathcal{T}$ contains X , so $\bigcap T$ is non-empty and is thus a subsemigroup. Indeed, it is the smallest subsemigroup of S that contains X . This subsemigroup, denoted by $\langle X \rangle$, is called the subsemigroup generated by X .

If $X \subseteq S$ is such that $\langle X \rangle = S$, then X is a generating set for S and X generates S . If there is a finite generating set for S , then S is said to be finitely generated.

Proposition 2.1.4. *Let $X \subseteq S$. Then*

$$\langle X \rangle = \{x_1x_2 \cdots x_n : n \in \mathbb{N}, x_i \in X\}.$$

2 The Semigroup

Proof. Let

$$U = \{x_1 x_2 \cdots x_n : n \in \mathbb{N}, x_i \in X\}.$$

Then U is closed under multiplication and so is a subsemigroup of S . Furthermore, $X \subseteq U$. Hence U must be one of the T_i in T , and so $\langle X \rangle \subseteq U$. Since $X \subseteq \langle X \rangle$ and $\langle X \rangle$ is closed under multiplication, we have $U \subseteq \langle X \rangle$. Therefore $\langle X \rangle = U$. $\square$

Given a subset X of a monoid M , we can also define the submonoid generated by X . This submonoid is the intersection of all submonoids of M containing whose identities are 1_M , that is, the collection of submonoids of M containing X and 1_M . The intersection of the submonoids in M is non-empty and thus a submonoid. This is the smallest submonoid of M with identity 1_M that contains X , denoted by $\text{Mon}(X)$.

Definition 2.1.18. This submonoid $\text{Mon}(X)$ is called the submonoid generated by X .

Proposition 2.1.5. Let $X \subseteq M$. Then

$$\text{Mon}(X) = \{1_M x_1 x_2 \cdots x_n : n \in \mathbb{N} \cup \{0\}, x_i \in X\}.$$

Remark 2.1.19. Essentially, when we generate a submonoid of a monoid, we always include the identity of the monoid. If $X \subseteq M$ is such that $\text{Mon}(X) = M$, then X is a monoid generating set for M and M is generated by X as a monoid. Notice that if X is a generating set for M , then X is also a monoid generating set; on the other hand, if X is a monoid generating set for M , then $X \cup \{1_M\}$ is a generating set for M . Thus M is finitely generated if and only if there is a finite monoid generating set for M .

2.1.5 Ideals and Monogenic Semigroup

Definition 2.1.20 (Ideal). Let T be a subset of S . The subset T is a left ideal of S if it is closed under left multiplication by any element of S ; that is, if $ST \subseteq T$. It is a right ideal of S if it is closed under right multiplication by any element of S ; that is, if $TS \subseteq T$. It is a two-sided ideal, or simply an ideal, of S if it is closed under both left and right multiplication by elements of S ; that is, if $ST \cup TS \subseteq T$.

Every ideal, whether left, right, or two-sided, is a subsemigroup.

Definition 2.1.21 (Principal ideal). For any $x \in S$, define

$$L(x) = S^1 x = \{x\} \cup Sx, \quad R(x) = x S^1 = \{x\} \cup xS, \quad J(x) = S^1 x S^1 = \{x\} \cup xS \cup Sx \cup SxS.$$

Then $L(x)$, $R(x)$, and $J(x)$ are, respectively, the principal left ideal generated by x , the principal right ideal generated by x , and the principal ideal generated by x . As they are, respectively, a left ideal, a right ideal, and a (two-sided) ideal.

Example 2.1.22. 1. Consider the semigroup $(\mathbb{N}, +)$. Let $n \in \mathbb{N}$ and let

$$I_n = \{m \in \mathbb{N} : m \geq n\}.$$

Then I_n is an ideal of $\mathbb{N}$; indeed, $I_n = L(n) = R(n) = J(n)$.

2. Let S be a right zero semigroup. Let T be a subset of S . Then $ST = T$ since $xy = y$ for any $x \in S$ and $y \in T$. So T is a left ideal of S . On the other hand $TS = S$, and so T is a right ideal of S if and only if $T = S$.

3. Let G be a group. Let T be a subset of G . For any $x \in G$ and $y \in T$, we have $x = xy^{-1}y \in Gy$; hence $Gy = G$. So T is a left ideal if and only if $T = G$; similarly, T is a right ideal if and only if $T = G$. So the only left ideal and right ideal of G is G itself.

Definition 2.1.23 (Rees congruence and Rees quotient). Let S be a semigroup and I a proper ideal of S . The *Rees congruence* on S modulo I is

$$\rho_I = (I \times I) \cup 1_S,$$

so that $x \rho_I y$ if and only if $x = y$ or $x, y \in I$. The quotient semigroup S/ρ_I , also written as S/I and called the *Rees quotient* of S by I , may be identified with

$$S/I = \{I\} \cup \{\{x\} : x \in S \setminus I\},$$

which can be regarded as I together with the elements of $S \setminus I$.

In S/I the product of two elements in $S \setminus I$ is same as their product in S if this lies in $S \setminus I$; otherwise the product is I .

Since the element I is the zero element of the semigroup, another way of thinking S/ρ_I is as $(S \setminus I) \cup \{0\}$, where all products not falling in $S \setminus I$ are zero.

The map $\varphi : S \rightarrow S/I$ given by $x\varphi = x$ if $x \notin I$ and $x\varphi = 0$ if $x \in I$ is a surjective homomorphism, called the *natural homomorphism*.

Definition 2.1.24. Let $\varphi : S \rightarrow T$ be a homomorphism. Then, φ is said to be a *rees morphism* if $\ker \varphi$ is a rees congruence on S .

Definition 2.1.25 (Ideal extension). Let M be a semigroup and T a semigroup with zero 0 . A semigroup E is called an *ideal extension* of M by T if M is an ideal of E and the Rees quotient semigroup E/M is isomorphic to T .

Definition 2.1.26 (Monogenic semigroup). Suppose S is generated by a single element x ; that is, $S = \langle x \rangle$. Then by Proposition 2.1.4,

$$S = \{x^n : n \in \mathbb{N}\},$$

and we call S the monogenic semigroup generated by x .

There are two possibilities: either all the powers of x are distinct, or there exist $k, m \in \mathbb{N}$ such that $x^k = x^{k+m}$. In the latter case, x is said to be periodic. If k and m are minimal, then S is finite, with $k + m - 1$ elements,

$$x, x^2, \dots, x^{k-1}, x^k, x^{k+1}, \dots, x^{k+m-1}.$$

We call k the *index* of x and m the *period* of x .

Remark 2.1.27. Let $K = \{x^k, x^{k+1}, \dots, x^{k+m-1}\}$. It is easy to see that K is an ideal of S . Furthermore, K is a subgroup of S , since the integers $k, k+1, \dots, k+m-1$ are representatives for congruence classes of the integers modulo m . In particular, some power of x is the identity of this subgroup and is therefore an idempotent.

Definition 2.1.28. A periodic semigroup is one in which every element is periodic.

2.1.6 Homomorphisms

Definition 2.1.29. Let S and T be semigroups. A map $\varphi : S \rightarrow T$ is a *homomorphism* if

$$(xy)\varphi = (x\varphi)(y\varphi) \quad \forall \quad x, y \in S.$$

If S and T are monoids, then φ is a *monoid homomorphism* if

$$(xy)\varphi = (x\varphi)(y\varphi) \quad \forall \quad x, y \in S \quad \text{and} \quad 1_S\varphi = 1_T.$$

A *monomorphism* is an injective homomorphism. An *epimorphism* is a surjective homomorphism. If $\varphi : S \rightarrow T$ is an epimorphism, then T is a homomorphic image of S .

Definition 2.1.30. An *isomorphism* is a bijective homomorphism.

It is easy to prove that a homomorphism $\varphi : S \rightarrow T$ is an isomorphism if and only if there is a homomorphism $\varphi^{-1} : T \rightarrow S$ such that $\varphi\varphi^{-1} = id_T$ and $\varphi^{-1}\varphi = id_S$. If there is an isomorphism $\varphi : S \rightarrow T$, then we say that S and T are isomorphic, and denote this by $S \cong T$.

It is easy to prove that if $\varphi : S \rightarrow T$ is a homomorphism and S' and T' are subsemigroups of S and T , respectively, then $S'\varphi$ is a subsemigroup of T and $T'\varphi^{-1}$ is a subsemigroup of S . In particular, putting $S' = S$ shows that $\text{im } \varphi$ is a subsemigroup of T . If φ is a monomorphism, then S is isomorphic to the subsemigroup $\text{im } \varphi$ of T .

Definition 2.1.31. The *kernel* of a homomorphism $\varphi : S \rightarrow T$ is the binary relation

$$\ker \varphi = \{(x, y) \in S \times S : x\varphi = y\varphi\}.$$

Notice that φ is a monomorphism if and only if $\ker \varphi$ is the identity relation (that is, $\ker \varphi = id_S$).

The following result shows that every semigroup is isomorphic to a subsemigroup of a transformation semigroup. This is the semigroup analogue of Cayley's theorem for groups, which states that every group is isomorphic to a subgroup of a symmetric group.

Theorem 2.1.6. For any $x \in S$, let $\rho_x \in \mathcal{T}_{S^1}$ be the map defined by $s\rho_x = sx$ for all $s \in S^1$. Then the map $\varphi : S \rightarrow \mathcal{T}_{S^1}$ given by $x \mapsto \rho_x$ is a monomorphism of S .

Proof. Let $x, y, s \in S$. Then

$$s\rho_x\rho_y = (s\rho_x)\rho_y = (sx)\rho_y = (sx)y = s(xy) = s\rho_{xy}.$$

Hence $\rho_x\rho_y = \rho_{xy}$, and so $(xy)\varphi = (x\varphi)(y\varphi)$. Therefore φ is a homomorphism.

Furthermore, if $x\varphi = y\varphi$, then $\rho_x = \rho_y$. Hence

$$1\rho_x = 1\rho_y,$$

that is, $x = y$. Therefore φ is injective. □

A map $\varphi : S \rightarrow T$ is an *anti-homomorphism* if

$$(xy)\varphi = (y\varphi)(x\varphi)$$

for all $x, y \in S$.

Remark 2.1.32. An *endomorphism* is a homomorphism from a semigroup S to itself. The set of all endomorphisms on S is denoted $\text{End}(S)$ and forms a subsemigroup of $\mathcal{T}_S$.

A semigroup S is *group-embeddable* if there is a group G and a monomorphism $\varphi : S \rightarrow G$. In this case, S is isomorphic to the subsemigroup $\text{im } \varphi$ of G . Clearly any group-embeddable semigroup is cancellative, but we shall see that there exist cancellative semigroups that are not group-embeddable.

2.1.7 Subdirect products

Definition 2.1.33 (Projection maps). Let $\mathcal{S} = \{S_i : i \in I\}$ be a collection of semigroups. For each $j \in I$, there is a *projection map* from the direct product $\prod_{i \in I} S_i$ onto S_j , taking an element of the direct product to its j -th component:

$$\pi_j : \prod_{i \in I} S_i \rightarrow S_j, \quad x\pi_j = (j)x.$$

Notice that every π_j is an epimorphism.

Definition 2.1.34 (Subdirect product). A *subdirect product* of $\mathcal{S}$ is [a semigroup isomorphic to] a subsemigroup P of the direct product $\prod_{i \in I} S_i$ such that $P\pi_i = S_i$ for all $i \in I$.

Definition 2.1.35 (Separation by epimorphisms). Let S be a semigroup. A collection of epimorphisms $\Phi = \{\varphi_i : S \rightarrow S_i : i \in I\}$ is said to separate the elements of S if they have the property that

$$(\forall i \in I)(x\varphi_i = y\varphi_i) \implies x = y.$$

Proposition 2.1.7. A semigroup S is a subdirect product of a collection of semigroups $\mathcal{S} = \{S_i : i \in I\}$ if and only if there exists a collection of epimorphisms $\Phi = \{\varphi_i : S \rightarrow S_i : i \in I\}$ that separate the elements of S .

Proof. If S is a subdirect product of $\mathcal{S}$, then the collection of projection maps restricted to S (that is, $\{\pi_i|_S : S \rightarrow S_i : i \in I\}$) separates the elements of S .

On the other hand, suppose the collection Φ separates the elements of S . Define $\psi : S \rightarrow \prod_{i \in I} S_i$ by letting the i -th component of $s\psi$ be $s\varphi_i$; that is, $(i)(s\psi) = s\varphi_i$. Then ψ is a homomorphism since each φ_i is a homomorphism. Furthermore, $s\psi = t\psi$ implies $s\varphi_i = t\varphi_i$ for all $i \in I$, which means that Φ separates the elements of S . Hence ψ is injective, so S is isomorphic to the subsemigroup $\text{im } \psi$ of $\prod_{i \in I} S_i$. Finally, the projection maps π_i are all surjective since each φ_i is surjective. So $\text{im } \psi$ is a subdirect product of $\mathcal{S}$. $\square$

Proposition 2.1.8. Let S be a semigroup and let $\Sigma = \{\sigma_i : i \in I\}$ be a collection of congruences on S . Let $\sigma = \bigcap_{i \in I} \sigma_i$. Then S/σ is a subdirect product of $\{S/\sigma_i : i \in I\}$.

Proof. For each i , there is a homomorphism $\varphi_i : S/\sigma \rightarrow S/\sigma_i$ with $[x]_\sigma \varphi_i = [x]_{\sigma_i}$. These maps are well-defined by Lemma 1.1.13. Clearly, the homomorphisms φ_i are surjective. Furthermore, the collection $\Phi = \{\varphi_i : i \in I\}$ separates the elements of S/σ , since if $[x]_\sigma \varphi_i = [y]_\sigma \varphi_i$ for all $i \in I$, then $[x]_{\sigma_i} = [y]_{\sigma_i}$ for all $i \in I$, which implies $(x, y) \in \sigma$. Therefore S/σ is a subdirect product of $\{S/\sigma_i : i \in I\}$ by Proposition 2.1.7. $\square$

Proposition 2.1.9. Let M be a monoid and let E be an ideal extension of M by T . Then E is a subdirect product of M and T .

Proof. Let $\varphi : E \rightarrow T$ be the natural homomorphism $x\varphi = [x]_M$. Let $\psi : E \rightarrow M$ be given by $x\psi = x1_M$. Then

$$\begin{aligned} (x\psi)(y\psi) &= x1_M y1_M \\ &= xy1_M \\ &= (xy)\psi, \end{aligned}$$

since $y1_M$ lies in the ideal M of E . Thus ψ is a homomorphism. Both φ and ψ are clearly surjective. Furthermore, if $x\varphi = y\varphi$, then either $x, y \in E - M$ and $[x]_M = [y]_M$, so $x = y$, or $x, y \in M$ and $x1_M = y1_M$, so $x = y$. Thus the collection of epimorphisms $\{\varphi, \psi\}$ separates elements of E , and so E is a subdirect product of M and T . $\square$

2.1.8 Actions

Definition 2.1.36 (Semigroup action). A *semigroup action* of a semigroup S on a set A is an operation

$$\cdot : A \times S \rightarrow A$$

that is compatible with the semigroup multiplication, in the sense that

$$(a \cdot x) \cdot y = a \cdot (xy)$$

for all $a \in A$ and $x, y \in S$.

We call such a semigroup action an *action of S on A* , or an *S -action on A* , and say that S acts on A .

Example 2.1.37. 1. Any subsemigroup S of $\mathcal{T}(A)$ acts on A by $a \cdot \varphi = a\varphi$ (where $\varphi \in \mathcal{T}(A)$).

2. Let S be a subsemigroup of a semigroup T . Then S acts on T via $t \cdot x = tx$ for all $t \in T$ and $x \in S$. In particular, this holds when $T = S$ or when $T = S^1$.

Given an action, we can define a map $\varphi : S \rightarrow \mathcal{T}(A)$, where the transformation $s\varphi$ is such that $a(s\varphi) = a \cdot s$. The condition $(a \cdot x) \cdot y = a \cdot (xy)$ implies that φ is a homomorphism. Conversely, given a homomorphism $\varphi : S \rightarrow \mathcal{T}(A)$, we can define an action by $a \cdot s = a(s\varphi)$, which satisfies $(a \cdot x) \cdot y = a \cdot (xy)$. There is thus a one-to-one correspondence between actions of a semigroup S on A and homomorphisms $\varphi : S \rightarrow \mathcal{T}(A)$.

Definition 2.1.38 (Free action). An action of S on A is *free* if distinct elements of S act differently on every element of A , or, equivalently,

$$(\forall x, y \in S) \left((\exists a \in A) (a \cdot x = a \cdot y) \Rightarrow x = y \right).$$

Definition 2.1.39 (Transitive action). An action of S on A is *transitive* if for all $a, b \in A$, there exists some element $s \in S$ such that $a \cdot s = b$.

Definition 2.1.40 (Regular action). An action of S on A is *regular* if it is both free and transitive.

Suppose A is also a semigroup. An action of S on A is by endomorphisms if $s\varphi \in \text{End}(A)$ for each $s \in S$; in this case,

$$(ab) \cdot x = (a \cdot x)(b \cdot x)$$

for all $a, b \in A$ and $x \in S$.

Note 2.1.41. The above discussions concern a right semigroup action. There is a dual notion of a left semigroup action

Definition 2.1.42. A *left semigroup action* of S on A is an operation

$$\cdot : S \times A \rightarrow A$$

satisfying

$$s \cdot (t \cdot a) = (st) \cdot a.$$

This corresponds to a map $\varphi : S \rightarrow \mathcal{T}(A)$, where $a(s\varphi) = s \cdot a$. This map φ is an anti-homomorphism since

$$a(t\varphi)(s\varphi) = (t \cdot a)(s\varphi) = s \cdot (t \cdot a) = (st) \cdot a = a((st)\varphi).$$

The definitions of actions being free, transitive, regular, and by endomorphisms also apply to left actions.

Remark 2.1.43. The correspondence of right actions with homomorphisms and left actions with anti-homomorphisms depends on writing mappings on the right and composing them left to right. When mappings are written on the left and composed right to left, right actions correspond to anti-homomorphisms and left actions correspond to homomorphisms.

2.2 Free semigroups and presentations

Informally, the free semigroup on a set A is the unique most general semigroup generated by A , in the sense that every other semigroup generated by A is a homomorphic image (and hence a factor semigroup) of the free semigroup on A .

Definition 2.2.1. An alphabet A is an abstract set whose elements are called *letters*.

2.2.1 Free Semigroups

A word over A is a finite sequence

$$(a_1, a_2, \dots, a_n),$$

where $a_i \in A$. The length of this word is defined to be n .

The empty sequence is called the *empty word* and is denoted by ε ; its length is 0. We denote the set of all non-empty words over A by A^+ , and the set of all words, including the empty word, by A^* . That is

$$A^* = A^+ \cup \{\varepsilon\}.$$

The multiplication of words is defined by concatenation:

$$(a_1, \dots, a_n)(b_1, \dots, b_m) = (a_1, \dots, a_n, b_1, \dots, b_m).$$

Clearly, this multiplication is associative, and so A^+ forms a semigroup and A^* forms a monoid. The empty word ε is the identity of A^* .

Definition 2.2.2. The semigroup A^+ is called the *free semigroup* on A , and the monoid A^* as *free monoid* on A .

We shall usually write

$$a_1 a_2 \cdots a_n$$

instead of $(a_1, a_2, \dots, a_n)$, which is possible because of associativity. We denote the length of a word u by $|u|$. In particular,

$$|u| = 0 \iff u = \varepsilon.$$

Lemma 2.2.1. Let $u, v \in A^*$. Then

$$|uv| = |u| + |v|.$$

2.2.2 Results and Properties

Definition 2.2.3 (Equidivisible semigroup). A semigroup S is called *equidivisible* if, for all $x, y, z, t \in S$,

$$xy = zt$$

implies that either

$$(\exists p \in S)(x = zp \text{ and } t = py),$$

or

$$(\exists q \in S)(z = xq \text{ and } y = qt).$$

Every group is equidivisible. Indeed, if $xy = zt$, then one may take $p = z^{-1}x$ or $q = x^{-1}z$.

Proposition 2.2.2. *The free semigroup A^* is equidivisible.*

Proof. Suppose $x, y, z, t \in A^*$ are such that $xy = zt$. Let

$$xy = zt = a_1 \cdots a_n,$$

where $a_i \in A$. Then, by the definition of multiplication in A^* , we have

$$x = a_1 \cdots a_k, \quad y = a_{k+1} \cdots a_n, \quad z = a_1 \cdots a_\ell, \quad t = a_{\ell+1} \cdots a_n,$$

where $0 \leq k, \ell \leq n+1$. We allow the values 0 and $n+1$ and formally take subwords $a_i \cdots a_j$, where $j < i$, to mean the empty word ε .

If $k \leq \ell$, let

$$q = a_{k+1} \cdots a_\ell.$$

Then $z = xq$ and $y = qt$.

On the other hand, if $k \geq \ell$, let

$$p = a_{\ell+1} \cdots a_k.$$

Then $x = zp$ and $t = py$. □

Proposition 2.2.3. *Let $A = \{x, y\}$. Let $B = \{b_i : i \in \mathbb{N}\}$. Then A^* contains a submonoid isomorphic to B^* .*

Proof. Define a map $\varphi : B \rightarrow A^*$ by

$$b_i \varphi = xy^i x.$$

Since B^* is free on B , this map φ extends to a unique homomorphism, which we also denote by φ , from B^* to A^* .

Suppose, with the aim of obtaining a contradiction, that φ is not injective. Then there exist $u, v \in B^*$ with

$$u\varphi = v\varphi.$$

Suppose u and v begin with the same symbol b_i ; that is, $u = b_i u'$ and $v = b_i v'$. Then

$$(b_i \varphi)(u' \varphi) = (b_i \varphi)(v' \varphi)$$

and so $u' \varphi = v' \varphi$ by cancellativity in A^* . Thus we can replace u by u' and v by v' and repeat this process until we have words u and v beginning with different symbols. Therefore assume that u and v begin with symbols b_i and b_j , respectively, where $i \neq j$; that is,

$$u = b_i u', \quad v = b_j v'.$$

Then

$$xy^i x(u' \varphi) = (b_i \varphi)(u' \varphi) = (b_j \varphi)(v' \varphi) = xy^j x(v' \varphi).$$

Assume $i > j$; the other case is similar. By cancellativity in A^* , we have

$$y^{i-j} x(u' \varphi) = x(v' \varphi),$$

which is a contradiction since $i - j > 0$. Therefore φ is injective and so B^* is isomorphic to $\text{im } \varphi$. $\square$

A similar result holds for free semigroups.

Definition 2.2.4 (Indecomposable element). An element a of a semigroup S is said to be *indecomposable* if there do not exist $u, v \in S$ such that

$$a = uv.$$

Proposition 2.2.4. Let M be a submonoid of A^* , and let

$$N = M \setminus \{\varepsilon\}.$$

Then $N \setminus N^2$ is the unique minimal generating set for M .

Proof. Clearly, any generating set for M must contain $N - N^2$. So we must show that

$$\langle N - N^2 \rangle = M.$$

Clearly,

$$\langle N - N^2 \rangle \subseteq M;$$

we have to prove that

$$M \subseteq \langle N - N^2 \rangle.$$

We already know that $\varepsilon \in \langle N - N^2 \rangle$, so it remains to show that

$$N \subseteq \langle N - N^2 \rangle.$$

We proceed by induction on the length of elements of N . Let

$$k = \min\{|u| : u \in N\}.$$

Then if $|u| = k$ and $u \in N$, then by Lemma 2.2.1, u cannot be factored as a product of two elements of N . So

$$u \in N - N^2 \subseteq \langle N - N^2 \rangle.$$

Thus all words of length k in N lie in $\langle N - N^2 \rangle$.

So assume that all words of length less than l in N lie in $\langle N - N^2 \rangle$. Let $u \in N$ with $|u| = l$. If $u \in N - N^2$, then

$$u \in \langle N - N^2 \rangle.$$

On the other hand, if $u \notin N - N^2$, then $u \in N^2$ and so

$$u = u' u''$$

for $u', u'' \in N$. By Lemma 2.2.1,

$$|u'|, |u''| < |u| = l.$$

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So, by assumption,

$$u', u'' \in \langle N - N^2 \rangle$$

and hence

$$u \in \langle N - N^2 \rangle.$$

Therefore, by induction,

$$N \subseteq \langle N - N^2 \rangle.$$

Hence

$$M = \langle N - N^2 \rangle.$$

Therefore $N - N^2$ is a generating set for M , and since every generating set must contain $N - N^2$, it is the unique minimal generating set. $\square$

The elements of $N \setminus N^2$ are precisely the *indecomposable* elements of M .

Definition 2.2.5. The base of a submonoid or a subsemigroup M of A^* is defined to be $N - N^2$, where $N = M - \{\varepsilon\}$.

The base is the unique minimal generating set for M .

Corollary 2.2.5. The alphabet A is the base of both A^+ and A^* .

Proposition 2.2.6. Let $u, v \in A^*$. Then

$$uv = vu$$

if and only if there exist $w \in A^*$ and $i, j \in \mathbb{N}$ such that

$$u = w^i \quad \text{and} \quad v = w^j.$$

Proof. If $u = w^i$ and $v = w^j$, then

$$uv = w^{i+j} = vu.$$

To prove the other direction, we proceed by induction on $|uv|$. If

$$|uv| \leq 2,$$

then clearly $uv = vu$ implies $u = v$. So assume the result holds for $|uv| < k$. Suppose $uv = vu$. Interchanging u and v if necessary, assume

$$|u| \geq |v|.$$

By Proposition 2.2.2, there exists $p \in A^*$ such that

$$u = vp \quad \text{and} \quad u = pv.$$

If $p = \varepsilon$, then $u = v$. Otherwise,

$$vp = pv,$$

where

$$|vp| < |uv|.$$

So by induction,

$$v = w^j \quad \text{and} \quad p = w^i$$

for some $w \in A^*$ and $i, j \in \mathbb{N}$. Thus

$$u = vp = w^{i+j}.$$

Hence, by induction, the result holds for all $u, v \in A^+$. $\square$

Proposition 2.2.7. *A semigroup S is free if and only if every element of S has a unique representative as a product of elements of $S - S^2$.*

Proof. Clearly every element of A^+ has a unique representative as a product of elements of A . So suppose every element of S has a unique representative as a product of elements of $A = S - S^2$.

We will show that S satisfies the definition of freedom. Let T be a semigroup and $\varphi : A \rightarrow T$ a map. Define a map $\varphi^+ : S \rightarrow T$ by letting

$$s\varphi^+ = (a_1\varphi)(a_2\varphi) \cdots (a_n\varphi),$$

where $a_1a_2 \cdots a_n$ is the unique representative of s as a product of elements $a_i \in A$.

Notice that if $t \in S$ is uniquely represented by $b_1 \cdots b_m$, where $b_i \in A$, then st has unique representative

$$a_1 \cdots a_nb_1 \cdots b_m.$$

Hence φ^+ is a homomorphism. It is clear that φ^+ is the unique homomorphism extending φ , and so S is free on A . $\square$

Example 2.2.6. Let $A = \{x\}$ and let $S = \langle x^2, x^3 \rangle$.

Then

$$S - S^2 = \{x^2, x^3\}.$$

But $x^5 \in S$ and

$$x^5 = x^2x^3 = x^3x^2,$$

so x^5 has two distinct representatives as a product of elements of $S - S^2$. Hence S is not free.

The above example shows that a free semigroup contains subsemigroups which are not free. In contrast, every subgroup of a free group is free by the Nielsen-Schreier theorem.

Definition 2.2.7 (Length function). A *length function* for a semigroup S is a homomorphism $s \mapsto |s|$. A proper length function for a monoid M is a length function with the additional property that

$$|s| = 0 \implies s = 1_M.$$

Notice that the usual notion of length for free semigroups as defined previously is also a proper length function.

Proposition 2.2.8. *A monoid M is equidivisible and has a proper length function if and only if it is isomorphic to A^* .*

Proof. We have already seen that A^* is equidivisible and has a proper length function. So suppose that M is an equidivisible monoid with a proper length function. Let

$$N = M - \{1_M\}.$$

Notice that N is a subsemigroup since

$$x, y \in N \implies (|x| \neq 0 \wedge |y| \neq 0) \implies |xy| = |x| + |y| \neq 0 \implies xy \neq 1_M.$$

Let $A = N - N^2$ and $w \in N^2$. Then $w = w'w''$ with

$$0 < |w'| < |w| \quad \text{and} \quad 0 < |w''| < |w|.$$

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Continuing this factoring process, we get

$$w = a_1 \cdots a_n,$$

where $a_i \in A$. So every element of M has a representative as a product of elements of A . Suppose w has another such representative $b_1 \cdots b_m$. We aim to show by induction on $m + n$ that the two representatives are identical. If $m + n = 2$, then

$$a_1 = w = b_1.$$

So suppose the result holds for all $m + n < k$.

Since

$$a_1 \cdots a_n = b_1 \cdots b_m,$$

equidivisibility shows that there exists $p \in M$ with either

$$a_1 = b_1 p \quad \text{and} \quad p a_2 \cdots a_n = b_2 \cdots b_m,$$

or

$$b_1 = a_1 p \quad \text{and} \quad p b_2 \cdots b_m = a_2 \cdots a_n.$$

Since $a_1 \in A = N - N^2$, it cannot be written as a product of two non-identity elements. Thus in the first case $p = 1_M$, and so

$$a_1 = b_1, \quad a_2 \cdots a_n = b_2 \cdots b_m.$$

The second case is similar. Applying the induction hypothesis shows

$$m = n$$

and

$$a_i = b_i$$

for all $i \geq 2$.

Thus every element of N has a unique representative as a product of elements of A , and so $N \cong A^+$ by Proposition 2.2.7. Hence

$$M = N \cup \{1_M\} \cong A^*.$$

□

Definition 2.2.8. Let x be an element of a monoid M . A non-trivial left factor of x is an element y such that $x = yz$ for some $z \in M - \{1_M\}$.

Proposition 2.2.9. A monoid M is free if it has the following four properties:

1. the only left- and right-invertible element of M is 1_M ;
2. M is cancellative;
3. M is equidivisible;
4. no element of M has infinitely many distinct non-trivial left factors.

Proof. Clearly, A^* has the four properties. So suppose M has the four properties (1)–(4). We first show that $N = M - \{1_M\}$ is a subsemigroup of M .

Let $x, y \in M$ with $xy = 1_M$. Then

$$(yx)^2 = y(xy)x = yx.$$

So yx is an idempotent. Since M is cancellative, $yx = 1_M$. Thus x and y are left and right inverses of each other, and hence

$$x = y = 1_M.$$

Therefore $x, y \in N$ implies $xy \in N$, and so N is a subsemigroup.

Now suppose

$$N = N^2.$$

Pick $x \in N$. Then there exists some m such that x has exactly m distinct non-trivial left factors. But since

$$N^{m+2} = N,$$

we can write

$$x = y_1 \cdots y_{m+2}$$

for $y_i \in N$. The products

$$y_1 \cdots y_k, \quad k < m + 2,$$

are all non-trivial left factors of x . Furthermore, they are all distinct, for if

$$y_1 \cdots y_k = y_1 \cdots y_{k+1},$$

then cancellativity shows that

$$y_{k+1} \cdots y_{m+2} = 1_M,$$

which is impossible since N is a subsemigroup. Thus x has at least $m + 1$ distinct non-trivial left factors, which is a contradiction. Hence $N^2 \subsetneq N$.

Thus $N - N^2$ is non-empty.

Similarly, suppose

$$x \in \bigcap \{N^m : m \in \mathbb{N}\}.$$

Suppose x has exactly m distinct non-trivial left factors. Since $x \in N^{m+2}$, the same contradiction arises. So $\bigcap \{N^m : m \in \mathbb{N}\}$ is empty. Therefore

$$N \supsetneq N^2 \supsetneq N^3 \supsetneq \cdots,$$

and hence every element $x \in N$ lies in $N^m - N^{m+1}$ for some unique m , and so x has an expression

$$x = n_1 \cdots n_m$$

as a product of elements $n_i \in N$.

Arguing by equidivisibility as in the proof of Proposition 2.2.8, we see that each element has a unique expression of this form. So M is free by Proposition 2.2.7. $\square$

2.2.3 Universal Property

Let F be a semigroup and let A be an alphabet. Let $\iota : A \hookrightarrow F$ be an embedding of A into F . Then the semigroup F is *free on A* if, for any semigroup S and map $\varphi : A \rightarrow S$, there is a unique homomorphism $\varphi^+ : F \rightarrow S$ extending φ . That is,

$$\iota\varphi^+ = \varphi,$$

or, equivalently, the following diagram commutes:

$$\begin{array}{ccc} A & \xrightarrow{\iota} & F \\ & \searrow \varphi & \downarrow \varphi^+ \\ & & S \end{array}$$

Proposition 2.2.10 (Uniqueness of free semigroups). *Let A be an alphabet and let F be a semigroup. Then F is free on A if and only if $F \cong A^+$.*

Proof. Suppose $F \cong A^+$. To show F is free on A , it is sufficient to show that A^+ is free on A . Let

$$\iota : A \rightarrow A^+$$

be the embedding map. Let S be a semigroup and $\varphi : A \rightarrow S$ be a map. Define

$$\varphi^+ : A^+ \rightarrow S$$

by

$$(a_1 a_2 \cdots a_n)\varphi^+ = (a_1\varphi)(a_2\varphi) \cdots (a_n\varphi).$$

It is easy to see that φ^+ is a homomorphism and that $\iota\varphi^+ = \varphi$.

We now have to prove that φ^+ is unique. So let

$$\psi : A^+ \rightarrow S$$

be an arbitrary homomorphism with $\iota\psi = \varphi$.

For any $a_1 a_2 \cdots a_n \in A^+$,

$$\begin{aligned} (a_1 a_2 \cdots a_n)\psi &= (a_1\psi)(a_2\psi) \cdots (a_n\psi) && \text{[since } \psi \text{ is a homomorphism]} \\ &= (a_1\varphi^+)(a_2\varphi^+) \cdots (a_n\varphi^+) && \text{[since } \iota\psi = \varphi = \iota\varphi^+] \\ &= (a_1 a_2 \cdots a_n)\varphi^+ && \text{[since } \varphi^+ \text{ is a homomorphism].} \end{aligned}$$

Thus $\psi = \varphi^+$. Hence φ^+ is the unique homomorphism from A^+ to S with $\iota\varphi^+ = \varphi$, and so A^+ is free on A .

Now let F be a semigroup that is free on A . Let

$$\iota_1 : A \rightarrow A^+ \quad \text{and} \quad \iota_2 : A \rightarrow F$$

be the embedding maps. Put $\varphi = \iota_2$ and $S = F$ in the definition of F being free on A to see that there is a homomorphism $\iota_2^+ : A^+ \rightarrow F$ with $\iota_1\iota_2^+ = \iota_2$.

Similarly, since F is free on A , there is a homomorphism $\iota_1^+ : F \rightarrow A^+$ with $\iota_2\iota_1^+ = \iota_1$. Therefore

$$\iota_1 = \iota_2\iota_1^+ \quad \text{and} \quad \iota_2 = \iota_1\iota_2^+.$$

That is, the following diagrams commute:

$$\begin{array}{ccc}
 A & \xrightarrow{\iota_1} & A^+ \\
 & \searrow \iota_2 & \downarrow \iota_2^+ \\
 & & F
 \end{array}
 \qquad
 \begin{array}{ccc}
 A & \xrightarrow{\iota_2} & F \\
 & \searrow \iota_1 & \downarrow \iota_1^+ \\
 & & A^+
 \end{array}$$

Therefore, by the uniqueness requirement in the definition of freedom,

$$\iota_2^+ \iota_1^+ = id_{A^+} \quad \text{and} \quad \iota_1^+ \iota_2^+ = id_F.$$

Hence ι_1^+ and ι_2^+ are mutually inverse isomorphisms, and so

$$F \cong A^+.$$

□